EVERY CONSTRUCTIBLE SHEAF ON A QUASI-COMPACT QUASI-SEPARATED SCHEME EMBEDS INTO A FINITE PRODUCT OR SUM OF PUSHFORWARDS OF CONSTANT SHEAVES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the absolute Noetherian approximation of X, the descent of the constructible sheaf to an approximant through the colimit description, the appeal to the Noetherian case, and the transport back along the pullback using base change for finite pushforwards.

Editorial note. The label lemma-constructible-maps-into-constant-general names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement retains the author's own bibliography reference environment to SGA4 Exposee IX Proposition 2.14, delivered verbatim as part of the counted statement.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Editorial note. Because the counted statement carries the author reference environment (SGA4 Exposee IX, Proposition 2.14), the chapter preamble excerpt includes the native verbatim package and the definitions of the reference, slogan and history environments (native preamble lines 8-13).

Allowed standard prerequisites (editorial).

- Absolute Noetherian approximation of a quasi-compact quasi-separated scheme. Limits, the proposition presenting X as a limit of schemes of finite type over Spec(Z) with affine transition morphisms. Supplies the presentation X = lim X_t used to reduce to the Noetherian case.
- Constructible sheaves as a colimit over the approximants. Same chapter, the lemma identifying the category of constructible sheaves on X with the colimit of the corresponding categories on the X_t. Supplies that the given constructible sheaf is the pullback of one on some X_t.
- The Noetherian case and base change for finite pushforwards. Same chapter, the Noetherian form of the embedding statement and the lemma that formation of a finite pushforward commutes with base change. Supplies the embedding over X_t and its transport to X along the pullback.

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Source reference. Original reference 09YU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12552–12552. Prerequisite pointer invoked at proof line(s) 13068; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09Z6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 12921–12921. Prerequisite pointer invoked at proof line(s) 13074; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0959: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8292–8292. Prerequisite pointer invoked at proof line(s) 13087; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Prerequisite pointer invoked at proof line(s) 13065; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme.*

(1) *Let $\mathcal{F}$ be a constructible sheaf of sets on $X_{étale}$. There exist an injective map of sheaves*
$$\mathcal{F} \longrightarrow \prod\nolimits_{i=1,\ldots,n} f_{i,*}\underline{E_i}$$
where $f_i : Y_i \to X$ is a finite and finitely presented morphism and E_i is a finite set.

(2) *Let $\mathcal{F}$ be a constructible abelian sheaf on $X_{étale}$. There exist an injective map of abelian sheaves*
$$\mathcal{F} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} f_{i,*}\underline{M_i}$$
where $f_i : Y_i \to X$ is a finite and finitely presented morphism and M_i is a finite abelian group.

(3) *Let Λ be a Noetherian ring. Let $\mathcal{F}$ be a constructible sheaf of Λ-modules on $X_{étale}$. There exist an injective map of sheaves of modules*
$$\mathcal{F} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} f_{i,*}\underline{M_i}$$
where $f_i : Y_i \to X$ is a finite and finitely presented morphism and M_i is a finite Λ-module.

Proof. We will reduce this lemma to the Noetherian case by absolute Noetherian approximation. Namely, by Limits, Proposition 01ZA we can write $X = \lim_{t\in T} X_t$ with each X_t of finite type over $\mathrm{Spec}(\mathbf{Z})$ and with affine transition morphisms. By Lemma 09YU the category of constructible sheaves (of sets, abelian groups, or Λ-modules) on $X_{étale}$ is the colimit of the corresponding categories for X_t. Thus our constructible sheaf $\mathcal{F}$ is the pullback of a similar constructible sheaf $\mathcal{F}_t$ over X_t for some t. Then we apply the Noetherian case (Lemma 09Z6) to find an injection
$$\mathcal{F}_t \longrightarrow \prod\nolimits_{i=1,\ldots,n} f_{i,*}\underline{E_i} \quad \text{or} \quad \mathcal{F}_t \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} f_{i,*}\underline{M_i}$$
over X_t for some finite morphisms $f_i : Y_i \to X_t$. Since X_t is Noetherian the morphisms f_i are of finite presentation. Since pullback is exact and since formation of $f_{i,*}$ commutes with base change (Lemma 0959), we conclude. □